This is an extremely difficult question for the evolutionist to answer. Most serious attempts to answer it involve sophisticated mathematical reasoning. I am frankly going to evade it except to say one thing. This is that at least some of the difficulty that theorists have with explaining the evolution of sex results from the fact that they habitually think of the individual as trying to maximize the number of his genes that survive. In these terms, sex appears paradoxical because it is an 'inefficient' way for an individual to propagate her genes: each child has only 50 per cent of the individual's genes, the other 50 per cent being provided by the sexual partner. If only, like a greenfly, she would bud-off children who were exact replicas of herself, she would pass 100 per cent of her genes on to the next generation in the body of every child. This apparent paradox has driven some theorists to embrace group selectionism, since it is relatively easy to think of group-level advantages for sex. As W. F. Bodmer has succinctly put it, sex 'facilitates the accumulation in a single individual of advantageous mutations which arose separately in different individuals.'

But the paradox seems less paradoxical if we follow the argument of this book, and treat the individual as a survival machine built by a short-lived confederation of long-lived genes. 'Efficiency' from the whole individual's point of view is then seen to be irrelevant. Sexuality versus non-sexuality will be regarded as an attribute under single-gene control, just like blue eyes versus brown eyes. A gene 'for' sexuality manipulates all the other genes for its own selfish ends. So does a gene for crossing-over. There are even genes—called mutators—that manipulate the rates of copying errors in other genes. By definition, a copying error is to the disadvantage of the gene which is miscopied. But if it is to the advantage of the selfish mutator gene that induces it, the mutator can spread through the gene pool. Similarly, if crossing-over benefits a gene for crossing-over, that is a sufficient explanation for the existence of crossing-over. And if sexual, as opposed to non-sexual, reproduction benefits a gene for sexual reproduction, that is a sufficient explanation for the existence of sexual reproduction. Whether or not it benefits all the rest of an individual's genes is comparatively irrelevant. Seen from the selfish gene's point of view, sex is not so bizarre after all.

This comes perilously close to being a circular argument, since the existence of sexuality is a precondition for the whole chain of reasoning that leads to the gene being regarded as the unit of selection. I believe